

Glossary of Key Terms

BDS-SMC2 UAV Rescue System — Lab 7 BeiDou Short Message Communication

BDS-3 System & Protocol		
Term	Abbr.	Definition
BeiDou Navigation Satellite System	BDS / BDS-3	China's independently developed global navigation satellite system. BDS-3 achieved global operational status on 31 July 2020. Unlike GPS, GLONASS, or Galileo, BDS-3 includes a built-in two-way messaging capability called Short Message Communication (SMC).
Short Message Communication	SMC	The two-way messaging service built into BDS-3. Allows a user terminal to transmit a short data payload (text, coordinates, telemetry) to a ground control centre via satellite and receive a response — without any terrestrial network infrastructure.
Radio Determination Satellite Service	RDSS	The active two-way positioning and messaging protocol used by BDS. The user terminal transmits an uplink signal; the satellite relays it to the Ground Control Centre (GCC), which computes the user's position and returns it via downlink. Unlike passive GNSS (GPS), RDSS is bidirectional.
Regional Short Message Communication	RSMC	The high-capacity BDS-3 SMC service served by three Geostationary (GEO) satellites covering the Asia-Pacific region. Supports up to 14,000 bits ($\approx$ 1,000 Chinese characters) per message with $\leq$ 1-second round-trip latency. Used in this experiment.
Global Short Message Communication	GSMC	The global BDS-3 SMC service served by 14 Medium Earth Orbit (MEO) satellites. Provides worldwide coverage but is limited to 560 bits ($\approx$ 40 Chinese characters) per message — making payload efficiency critical for multi-field telemetry.
Ground Control Centre	GCC	The ground-based facility that processes RDSS uplink signals from user terminals. The GCC measures signal time-of-flight from two or more GEO satellites plus a Digital Elevation Model (DEM) to compute the terminal's 2D+altitude position, then returns it via the satellite downlink.
Geostationary Earth Orbit Satellite	GEO	A satellite positioned at approximately 35,786 km altitude that remains stationary relative to Earth. BDS-3 uses three GEO satellites (C59, C60, C61) for RSMC service. From Hangzhou, these appear in the southern sky at elevations of approximately 38–52°.
Medium Earth Orbit Satellite	MEO	Satellites orbiting at approximately 19,000–24,000 km altitude. BDS-3 uses 14 MEO satellites for GSMC global coverage. MEO satellites provide better geometric diversity than GEO but at lower per-satellite processing capacity.

\$CCTXM Command	AT Cmd	The NMEA-style AT command used to trigger a BDS-3 short message transmission. Format: \$CCTXM,,*\r\n. A destination ID of 0 routes the message to the GCC relay for forwarding to the web platform.
XOR Checksum	CS	A simple error-detection value computed by XOR-ing all characters between the \$ and * in an NMEA-style command. Used to detect transmission corruption between the ESP32 and the BDS module over the UART connection.

Encoding & Compression

Term	Abbr.	Definition
ASCII Encoding	ASCII	American Standard Code for Information Interchange. Represents each character as 8 bits. Default format for coordinate transmission: LAT:30.4196,LON:120.2977 uses approximately 208 bits — a 69% overhead compared to binary encoding of the same data.
Binary Fixed-Point Encoding	BIN	A compact coordinate representation where latitude and longitude are each multiplied by 10,000 and stored as signed 32-bit integers. Preserves 4 decimal place precision (± 11 m accuracy) in exactly 64 bits total — a 69% reduction vs ASCII.
Huffman Encoding	HUF	A lossless data compression algorithm that assigns shorter bit codes to more frequently occurring characters. In this project, a dynamic Huffman tree is built at runtime from the character frequency distribution of the telemetry string, achieving up to 77% reduction vs ASCII.
Payload	—	The actual data content carried inside a BDS-3 SMC message — the coordinate, telemetry fields, or encrypted bytes. Distinguished from the message overhead (command header, checksum, framing) which is transmitted but carries no information.
Bit Budget	—	The maximum number of bits available for the payload in a single BDS-3 SMC message. RSMC: 14,000 bits. GSMC: 560 bits. Efficient encoding maximises the information that can be transmitted within this fixed limit.
Fixed-Point Arithmetic	—	A method of representing decimal numbers as integers by scaling them by a fixed factor. Example: $30.4196^\circ \times 10,000 = 304,196$ stored as an integer. Avoids floating-point format overhead while preserving the required precision.
Telemetry	—	A structured set of measurement fields transmitted from the rescue terminal to the ground station. In this project: latitude, longitude, altitude, battery percentage, mode, flags, and timestamp — 7 fields total.

Message Fragmentation	—	The splitting of a payload that exceeds the single-message bit limit into multiple sequential messages. BDS-3 has no native fragment reassembly protocol — if any fragment is lost, the entire payload is unrecoverable.
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Security & Encryption

Term	Abbr.	Definition
AES-128-CBC	AES	Advanced Encryption Standard with a 128-bit key in Cipher Block Chaining mode. The encryption algorithm implemented in this project using the ESP32's built-in mbedTLS library. Encrypts the 8-byte binary coordinate payload into a 16-byte (128-bit) ciphertext block — adding exactly 64 bits of overhead.
mbedTLS	—	A lightweight cryptographic library embedded in the ESP32 SDK (via ESP-IDF). Provides AES-128/256, SHA, and RSA without requiring external libraries. Enables encryption directly on the microcontroller with no additional hardware.
Pre-Shared Key	PSK	A secret key known to both the sender (ESP32) and the receiver (Python decoder) before communication begins, established at device setup time. Eliminates the need for runtime key exchange but requires secure distribution at deployment.
Key Infrastructure / Key Generation Centre	KGC	A trusted third-party server that manages, generates, and distributes cryptographic keys. Required by SM9 IBE and some SM4 implementations. The proposed PSK solution eliminates the need for a KGC at runtime.
Identity-Based Encryption	IBE / SM9	A public-key cryptography scheme where the receiver's public key is derived from their identity (e.g. email address or ID number). Used by Jin et al. (2024) for BDS-3 SMC. Requires the sender to know the receiver's identity and a KGC to be online.
SM4	SM4	A Chinese national symmetric block cipher standard with 128-bit block and key size, analogous to AES. Proposed for BDS-3 SMC security by IEEE (2023). Requires prior key exchange between sender and receiver.
PKCS7 Padding	—	A padding scheme that extends a plaintext to a multiple of the cipher block size by appending bytes whose value equals the number of padding bytes added. Used here to extend the 8-byte coordinate payload to 16 bytes for AES-128 encryption.
Cold-Start Emergency	—	A scenario where a survivor must transmit an encrypted SOS with no prior configuration — no pre-established keys, no knowledge of the rescue operator's identity, no internet. The gap this project's AES-PSK solution specifically addresses.

S-band Downlink	—	The radio frequency band (~2492 MHz) used by BDS-3 GEO satellites to broadcast return messages to user terminals. Without encryption, any receiver tuned to this frequency can intercept and read transmitted coordinates in plaintext.
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Hardware & System

Term	Abbr.	Definition
ESP32 DevKitC V4	ESP32	A dual-core 240 MHz microcontroller with built-in WiFi and Bluetooth, manufactured by Espressif Systems. Used in this project to run the firmware, generate AT commands, and communicate with the BDS module via UART2 (GPIO 16 RX, GPIO 17 TX).
BDS EVBKIT V3	EVBKIT	The BeiDou-3 Short Message evaluation board used in this experiment. Contains the BDS-3 communication module, SIM card slot, RS232 interface, and DC power input. Connects to the ESP32 via an RS232-to-TTL adapter.
RS232-to-TTL Adapter	MAX232	A signal level converter board that translates between RS232 voltage levels ($\pm 12V$) used by the BDS module and TTL voltage levels (0/3.3V) used by the ESP32. Wiring: VCC→3.3V (red), GND→GND (black), RXD→GPIO17 (blue), TXD→GPIO16 (green).
UART2	UART	Universal Asynchronous Receiver-Transmitter — the serial communication protocol used between the ESP32 and the BDS module. Configured at 9600 baud, 8 data bits, no parity, 1 stop bit (SERIAL_8N1) on pins GPIO 16 and 17.
XCOM V2.0	XCOM	A serial port debugging assistant (serial monitor) used to view raw BDS module responses in real time. Set to 115200 baud to match the ESP32 Serial Monitor output. Displays \$RDTXA acknowledgements and error codes.
NMEA-0183	NMEA	A serial communication standard originally designed for marine navigation instruments. BDS modules output position data as NMEA sentences (e.g. \$GNGGA, \$GNRMC). The \$CCTXM command uses NMEA-style framing with a \$ prefix and *checksum suffix.
Line-of-Sight	LOS	An unobstructed direct path between the user terminal's antenna and the satellite. Required for RSMC transmission. Obstructions (buildings, trees, terrain) attenuate or block the L-band signal, reducing or preventing successful transmission.

Experiment & Analysis

Term	Abbr.	Definition
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End-to-End Latency	E2E	The total time elapsed from when the ESP32 sends the \$CCTXM command to when the decoded coordinate is available on the receiving system. Composed of: UART serialisation + BDS module processing + satellite uplink + GCC processing + satellite downlink + web platform relay.
Success Rate	SR	The proportion of BDS-3 SMC transmission attempts that result in a confirmed acknowledgement (\$RDTXA or 'OK') within the 30-second timeout window. Expressed as a percentage with a 95% confidence interval.
95% Confidence Interval	95% CI	A statistical range within which the true population success rate is expected to fall with 95% probability. Computed as: $p \pm 1.96 \times \sqrt{(p(1-p)/n)}$, where p is the observed success rate and n is the number of trials.
Compression Ratio	CR	The ratio of the original (ASCII) payload size to the compressed payload size. Example: 208 bits ASCII / 64 bits binary = 3.25× compression ratio, representing a 69% size reduction.
serial_logger.py	—	Python script that reads the ESP32 Serial Monitor output over COM13 and logs all messages with PC-side timestamps to a CSV file. Used to capture [T1] transmission markers and BDS module responses for latency analysis.
field_test_logger.py	—	Python script that auto-detects ESP32 [TX#] transmission events over serial and logs success/fail/latency for each attempt to gap3_field_test.csv. Supports timeout detection (30 s) and failure logging.
telemetry_compare.py	—	Python script that encodes the same 7-field telemetry struct in ASCII, Binary, Huffman, and AES-128 formats, measures bit sizes for each, and saves results to gap6_telemetry.csv for Gap 6 analysis.
aes_decrypt.py	—	Python script that decrypts AES-128-CBC ciphertext received from the ESP32, using the matching pre-shared key and IV (pycryptodome library). Verifies coordinate recovery within $\pm 0.0001^\circ$ tolerance.